

Basic Information

Experiment No: 02

Title: Program to Check Whether a Given Number is Prime Palindrome or Not

CO: CO1

Objective / Aim

To write a Java program to check whether the given number is both a **prime number** and a **palindrome**.

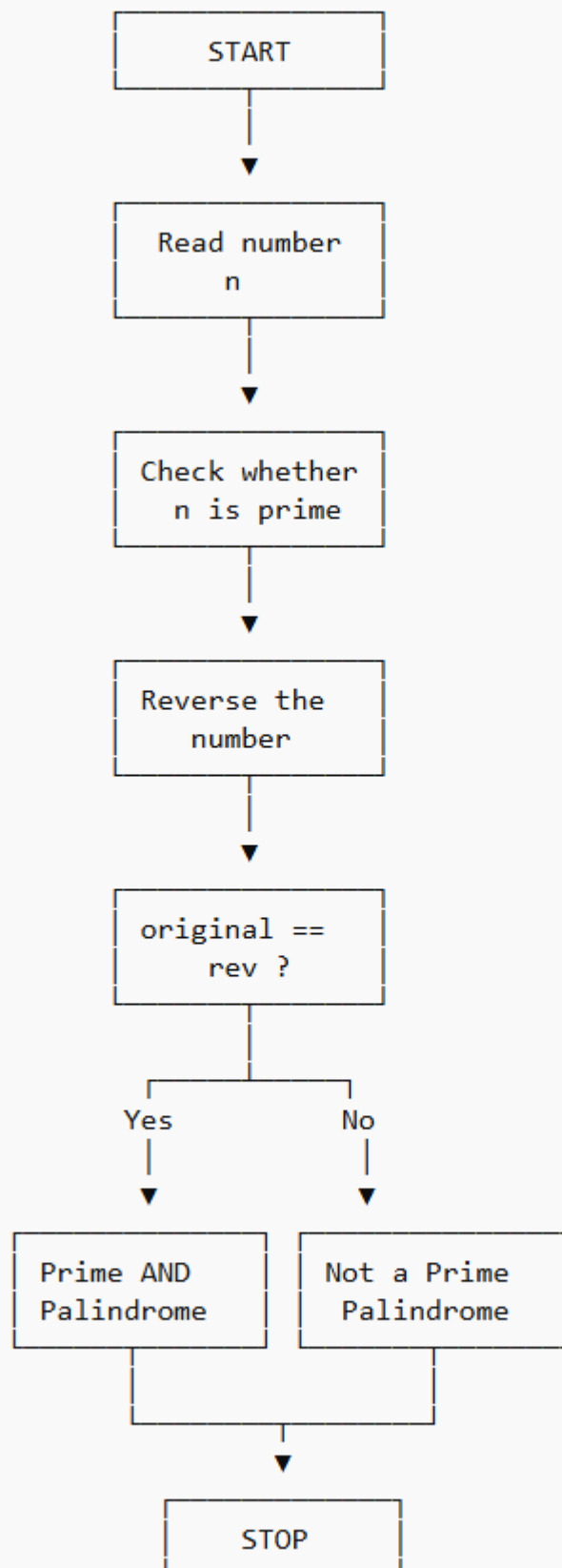
System Requirements

- **Hardware:** Standard PC with Core i3/i5 processor and 4GB/8GB RAM.
- **Software:** Windows/Linux/macOS.
- **Programming Language:** Java.
- **JDK:** JDK 17 or JDK 21.
- **IDE:** Eclipse, IntelliJ IDEA, NetBeans, or VS Code.

Algorithm / Logic Description

1. Import the Scanner class to accept input from the user.
2. Read the number n from the user.
3. Assume the number is prime by setting prime = true.
4. If the number is less than or equal to 1, set prime = false.
5. Check divisibility of the number from 2 to n/2.
6. If the number is divisible by any value, it is not prime.
7. Store the original number in the variable original.
8. Reverse the number using the % and / operators.
9. Compare the reversed number with the original number.
10. If the number is both prime and equal to its reverse, display "**prime palindrome**".
11. Otherwise, display "**not a prime palindrome**".
12. Close the Scanner object

UML DIAGRAM



```
import java.util.Scanner;
```

```
public class PrimePalindrome {
```

```
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        System.out.println("Enter a number to check: ");
```

```
        int n = sc.nextInt();
```

```
        // Checking for prime
```

```
        boolean prime = true;
```

```
        if (n <= 1) {
```

```
            prime = false;
```

```
        }
```

```
        for (int i = 2; i <= n / 2; i++) {
```

```
            if (n % i == 0) {
```

```
                prime = false;
```

```
                break;
```

```
            }
```

```
        }
```

```
        // Checking for palindrome
```

```
        int rev = 0, temp;
```

```
        int original = n;
```

```
        while (n > 0) {
```

```

        temp = n % 10;
        rev = rev * 10 + temp;
        n = n / 10;
    }

    // Checking both conditions
    if (prime && original == rev) {
        System.out.println("Given number " + original
            + " is a prime palindrome");
    } else {
        System.out.println("It is not a prime palindrome");
    }

    sc.close();
}
}

```

Compilation & Execution Commands

Save the file exactly as:

PrimePalindrome.java

Compile the program:

```
javac PrimePalindrome.java
```

Run the program:

```
java PrimePalindrome
```

Output Screen

Sample Output 1

Enter a number to check:

131

Given number 131 is a prime palindrome

Sample Output 2

Enter a number to check:

121

It is not a prime palindrome

Sample Output 3

Enter a number to check:

17

It is not a prime palindrome

Result

The Java program was successfully executed to determine whether the given number is **both prime and palindrome**.